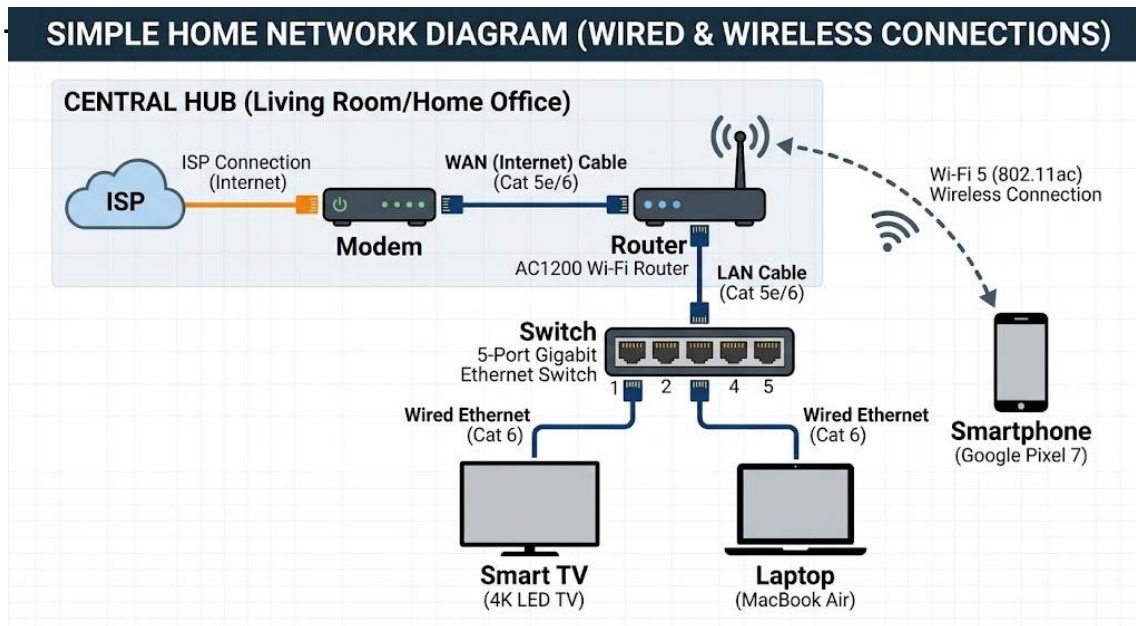


Network devices

Task – 1 Diagram of home router.



Task – 2 Network devices like router, lan etc..

Network Device	Primary Function	OSI Layer	Real-World Use Case
Router	Connects different networks together (e.g., your local home network to the internet) and	Layer 3	Streaming Netflix & YouTube: Directs incoming video streams from the internet across your local network specifically to your Smart TV or laptop, while assigning private IP addresses via DHCP.

	routes data packets using IP addresses.		
Switch	Connects multiple wired devices on the same local network (LAN) and intelligently forwards data frames directly to target MAC addresses.	Layer 2	Low-Latency LAN Gaming & Desktop PC Setup: Connects a gaming PC and a local NAS (Network Attached Storage) to the router over Ethernet, allowing fast file transfers and low-latency multiplayer gaming in games like <i>Valorant</i> or <i>Counter-Strike</i> .
Network Hub	Broadcasts all incoming data packets to every connected port, regardless of the destination device, sharing bandwidth across all nodes.	Layer 1	Legacy Packet Sniffing & Basic Diagnostics: Connecting a diagnostics laptop to monitor or capture all unencrypted network traffic sent between older legacy laboratory devices, though largely replaced by managed switches with port mirroring.
Wireless Access Point (WAP)	Transmits and receives wireless radio signals to bridge Wi-Fi devices onto a wired network segment.	Layer 2	Wi-Fi Coverage for Mobile Apps: Transmits Wi-Fi signal across a home or university dorm, enabling smart devices to scroll TikTok, send WhatsApp messages, or stream Spotify while moving between rooms without dropping the connection.

Task – 3 How data travels.

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Task – 4 Based on scenario.

- Uses your home's existing electrical wiring to transmit data packets between rooms without needing to run long Ethernet cables across floors. I plug the first powerline unit into a wall outlet near your Wi-Fi router and connect it using a short Ethernet cable. I plug the second unit into a wall outlet near your gaming PC and connect its Ethernet port directly to your PC. Data travels through the copper electrical lines from the router to your computer.